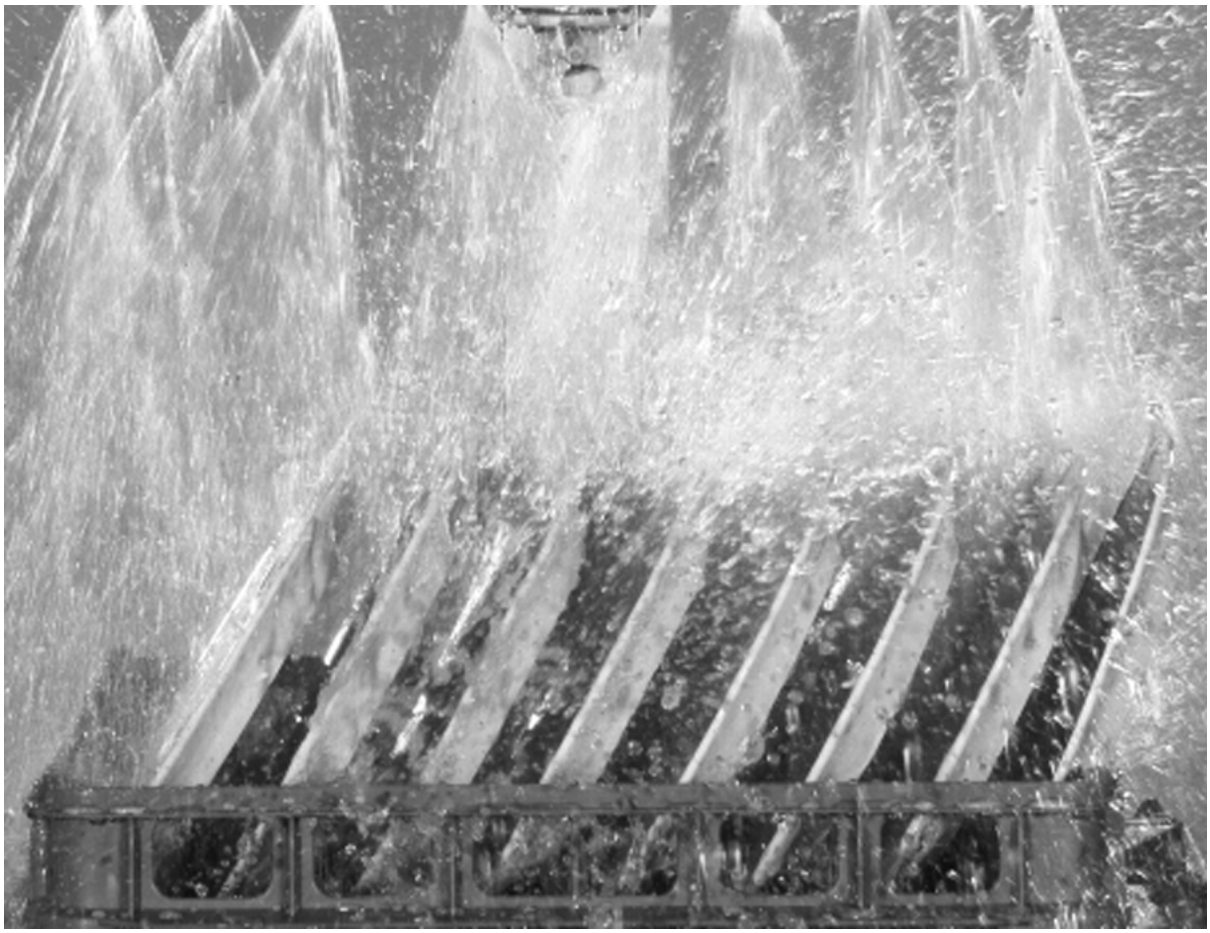
Operating instructions

Universal warewashing machine
FV 130.2
FV 250.2

Installation and commissioning instructions



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2 Declaration of conformity and other certificates

See separate EU declaration of conformity.

3 Transport, delivery, dimensions

3.1 Transport

- Observe transport instructions on the packing.
- The appliance must be moved with great care.
- Unpack the appliance.
- Check completeness of the delivery using the shipping documents.
- Examine the appliance for possible transit damage.
- Any damage must be reported immediately to the carrier, the insurance company and the manufacturer.

Damaged appliances must not be commissioned.

3.2 Dimensions, technical data, installation instructions

See attached technical sheet

4 General safety information

This model has exclusively been designed for the washing of dishes, cutlery and kitchen utensils. It must not be used for any other purpose.

For a trouble-free and safe operation of the appliance, the operating instructions must be carefully observed.

4.1 Explanation of the safety symbols

The following symbols are used in this booklet as important safety notes for the operator. The safety notes, especially the warnings, MUST be observed and followed at all times.

Warning: indicates possible hazard for persons, especially through electrical equipment.

Attention: indicates the endangering of system parts or possible functional damage.

No splashing water: prohibits the use of a high pressure hose.

Danger of explosion: indicates a potential explosion hazard.

Non-potable water: The water is not for drinking. Health can be endangered by drinking.

Danger of burning: indicates possible hazard due to hot surfaces or media.

Danger of crushing: indicates hazard caused by moving parts of the appliance.

Note: includes important additional information for the operator regarding the system or parts of the system and offers further information.

➢ **Instruction symbol:** placed in front of text requiring an action.

4.2 Operating conditions

It is taken for granted that the planning of the system, its installation, commissioning and maintenance are executed by properly instructed staff and that these works are checked by responsible specialists. The indications on the name plate of the machine must correspond to the technical sheet and the local connection conditions.

Conditions to be provided by the customer:

- ❑ Frost free storage and installation area
- ❑ Electrical connection in accordance with the technical sheet
- ❑ Fresh water connection in accordance with the technical sheet
- ❑ Waste water connection in accordance with the technical sheet

4.2.1 Requirements for the installation area

- ➢ Ensure that the storage and installation area is permanently frost free.

The machine is only frost-resistant in the state it is delivered or when provided with special features (option: frost drainage). If the appliance is installed in an area where the surrounding temperatures are below freezing point, damage can occur to components containing water, such as pump, solenoid valve, boiler, etc.

5 Installation, commissioning

5.1 Installation of the Universal warewashing machine

The enclosed technical sheet indicates the connected loads and consumption ratings of the appliance.

Small quantities of steam may escape from the door of the appliance. Furniture and equipment situated near the door must be protected.

An engineer from your local MEIKO Service Centre can install the appliance at the required position and connect the tables - upon request.

The following must be observed during the installation of the appliance:

- ➢ The complete unit must be levelled in both directions using a spirit level.
- ➢ Compensate for an uneven floor by adjusting the feet.
- ➢ Table joints must be sealed with detergent-resistant sealing compound (e.g. silicone).

5.2 Requirements for the electrical connection

The electrical connection of the appliance must only be made by a specialist as defined in DIN EN 50110-1.

The customer must guarantee the following points relating to the connection:

- The correct voltage and type of current must be available
- Mains supply lines must be protected according to regulations and provided with a main switch.
- Appliances must have a fixed connection and equi-potential bonding.
- If an unearthed neutral (N) is used with alternating current, the main switch must have 4-poles.
- When connecting to alternating current, a 5-pole terminal strip (L1, L2, L3, N, PE) must be used.
- If the electric supply is without neutral conductor (N), use a 4-pole terminal strip (L1, L2, L3, PE) when connecting to alternating current.
- Conductor colors: live conductor L1 = black/1, L2 = brown/2, L3 = black/3, neutral conductor N = blue/4, protective earthing conductor PE = green-yellow.

Protective measures as well as the connection of the equi-potential bonding must be carried out according to DIN VDE 0100-540 and conform to the local power utilities' regulations.

Do not use fuses to protect any additional consumers together with the universal wash machine.

- All conductor fixing screws must be re-tightened before commissioning the appliance.

The wiring diagram is behind the front panel or front cladding of the appliance. The enclosed wiring diagram must remain in the appliance.

5.3 Requirements for the fresh water connection

Each appliance carries the DVGW test symbol and does not require an extra safety valve in the water feed.

- Fresh water connection must be made according to DIN 1988 part 1-8 or EN 1717, DVGW-work sheet W 507, or according to local regulations.

The machine is equipped with a water supply air gap (group A, model A in accordance with DIN 1988 and EN 1717).

The flow pressure of the fresh water supply line must be at least 0.6 bar up to max. 5 bars (in front of the solenoid valve).

- If this flow pressure is not present, then increase the pressure by means of a booster pump or reduce it with a pressure reducing valve.
- A water stop is integrated into the fresh water inlet of the machine. This, together with the leak water switch in the floor pan of the base,

ensures that the fresh water supply will be cut off in the event of any leak

- Suitable protective measures must be taken to ensure that no iron particles can enter the appliance via the mains water supply. Similarly, precautions must be taken to prevent the entry of other metal particles, for example copper turnings. Corresponding instructions are contained in the installation drawing. Therefore suitable measures must be taken.
- A dirt trap must be fitted into the fresh water supply to protect the solenoid valve.

5.4 Requirements for the waste water connection

- A waste water pump is integrated in the waste water line. A siphon must be provided by the client on site (further information about this is in the technical sheet).
- A grease trap may be needed, depending on the machine application.

5.5 Correct operation

The machine must only be operated under the supervision of properly instructed staff.

Temperature of wash water = 58-60 °C.

Items that have been washed and components in contact with the wash water have the same temperature. Appropriate protective measures must be observed.

Open the door very carefully during the programme cycle, as otherwise wash water could splash out.

5.5.1 Emergency switch

- Set the local main switch to ”OFF“ or disconnect the local main fuse.

5.6 Authorised users / operators

It is assumed that two groups of users will come into contact with the appliance:

1. Service engineers from the manufacturer or the agency or installation engineers from the authorised dealer: these are qualified technicians and have good product knowledge.
2. Operators - with basic knowledge (our customers must instruct the operators).

Customer’s responsibility relating to the groups of users

- ❑ The persons responsible for safety must fully guarantee that only qualified persons carry out connection, adjustment or maintenance works on the open appliance.
- ❑ The permission for carrying out this type of work is given by the person responsible for the safety of staff and of the appliance to persons qualified because of their training, education, experience or instructions, or because of their knowledge regarding standards, regulations, accident-prevention regulations and the circumstances of the installation.
- ❑ The persons responsible for safety must instruct the operators with regard to the operation and the safety aspects of the appliance.

- ❑ Steps must be taken to ensure that the operating instructions supplied are available for both user groups. Service personnel must also have the documentation relating to the order available for them when undertaking any work and observe these instructions in order to avoid any hazard and/or damages.

5.7 Chemicals for the operation of the appliance

Only detergents and rinse-aids suitable for the use in industrial dishwashers may be used. Relevant information is provided by the manufacturers of these products.

If unsuitable products are used, the life expectancy of the dosing units can be reduced considerably.

Observe the dosing instructions of the manufacturer.

Detergents and rinse-aids can present a health hazard if they are not correctly used. Please observe the manufacturers' instructions on the original packing and on the safety data sheets.

If a de-scaling agent is used, observe the manufacturer's instructions strictly regarding handling and safety. After such an agent has been used the product must be completely removed from the machine, as even small residues are sufficient to destroy plastic parts and sealing materials.

5.8 Environmentally acceptable measures, measures for disposal

Each discarded appliance is to be made immediately unserviceable in order to avoid accidents at a later date.

- ➢ Therefore, set the local main switch to "OFF" disconnect the local main fuse.

The appliance can now be properly recycled (stainless steel, plastic materials, etc. separately).

6 Basic information on the universal appliance

Each machine is manufactured according to state of the art technology. Operation is safe.

Dangers could arise from this model if it is incorrectly operated by unsuitable operating staff or if it is not used for the purpose intended.

Liability

We accept no responsibility for damage to the appliance and other objects caused by operating faults or by failure to observe the operating instructions. Any modifications to the appliance - especially internal technical modifications - undertaken by unauthorised persons without the written permission of the manufacturer will invalidate the warranty and all product liability.

6.1 General description of the model

6.1.1 Type

Square basket appliance with stationary basket.

6.1.2 Wash principle

The appliance has one wash and one final rinse cycle.

The temperature regulator maintains the wash temperature. A centrifugal pump moves the water from the wash tank into the wash nozzles. The water jets hit the items to be washed from different directions. This guarantees an even wash.

The washing cycle is followed by the fresh water final rinse. The items are rinsed via a separate nozzle system with hot fresh water at 80 - 83° C. This also heats the items for the subsequent drying process. At the same time the final rinse water is used to regenerate the wash water, thus reducing the level of contamination of the wash water.

6.1.3 Detergent dosing (optional)

The detergent dosing unit is designed for the automatic addition of liquid alkaline detergent to the wash water.

The detergent is moved from the container into the wash tank by means of a hose line. The dosing unit is self-priming. Dosing takes place during each filling cycle and at the beginning of each programme cycle and is controlled by an automatic timer.

A dose of approx. 2 ml of detergent per litre of tank water is the correct normal concentration. This can be increased/reduced to 5 ml/l or to1 ml/l depending on the water quality, the items to be washed and the degree of soiling

6.1.4 Rinse aid dosing unit

The rinse aid dosing unit is designed to automatically add liquid final rinse aid into the fresh water.

The rinse aid is moved from the container into the fresh water supply line by means of a hose line. The dosing unit is self-priming. The dosing takes place during each filling cycle.

The correct dosing results in a smooth, even water film.

In case of overdosing, there are bubble and stripe formations - reduce dosing.

In case of under-dosing, water drops remain on the washed items - increase dosing.

The instructions for operating and adjustment of the dosing units can be found in the ”Service Instructions“.

6.2 Improper use

Universal warewashing machine and especially switch boxes and other electrical-technical components must not be sprayed with a water hose or a high-pressure hose.

Do not place any solvents or other easily flammable substances in the washing-up area, as this increases the danger from explosions.

The water in the washing-up area is not drinking water! Do not use the water from the washing-up area for food preparation or drinking!

Do not place anything on the open door of the appliance, especially heavy loads, as the machine could tip over!

The door must be opened very carefully during the wash cycle to reduce the risk of wash water splashing out of the appliance.

Steel pan scrubbers must not be used for the pre-scouring or for cleaning the items to be washed.

Do not wash any metal items which are not made of stainless steel in the machine.

The entry of metal particles (especially iron, tinplate, copper) MUST be prevented.

The appliance must not be used to transfer waste water from other sources into the drain (warning: risk of corrosion and blockage).

Only use suitable products for cleaning the stainless steel surfaces, which do not attack the material, form any deposits, nor cause any discolorations.

6.3 Emissions

- ❑ Work place noise level $L_{pA} \leq 70$ dB
- ❑ Water vapour

Small quantities of steam may escape from the door of the appliance. Furniture and equipment situated near the door must be protected.

6.4 Data relating to the electrical and hydraulic equipment

Technical data: installation drawing

7 Documentation

Installation drawing / technical sheet
Wiring diagram / Programming instructions

The Service Instructions are part of the Operating Instructions and should be kept in the appliance together with the wiring diagram.

8 Appendix

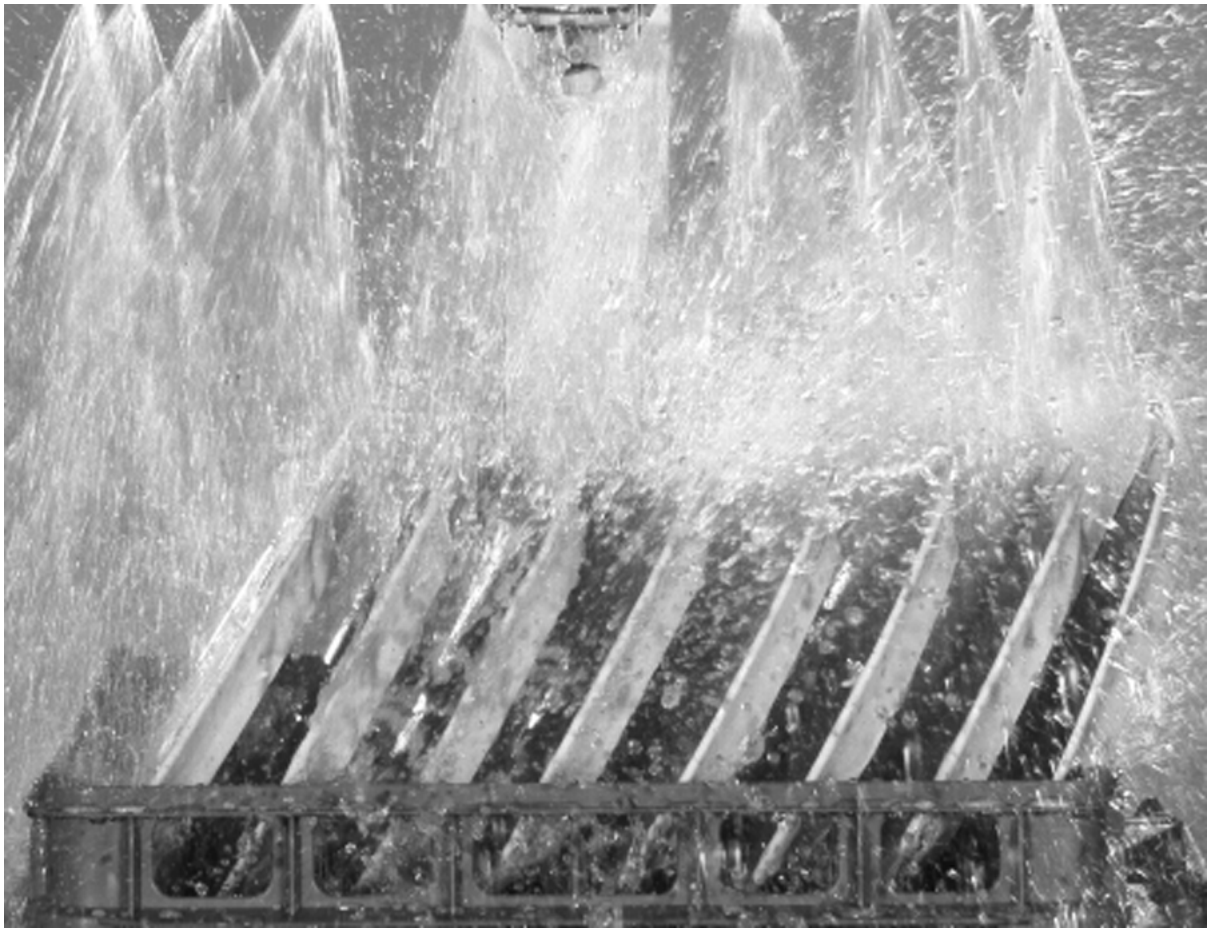
8.1 Name and address of manufacturer

MEIKO Maschinenbau GmbH & Co. KG
Englerstrasse 3
D-77652 Offenburg
Phone no.: (++ 49) 781 / 203 - 0
Fax no.: (++ 49) 781 / 203 - 1121
www.meiko.de
e-mail: info@meiko.de

Operating instructions

Universal warewashing machine
FV 130.2
FV 250.2

Operating instructions



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2 Operation

This appliance is a technically-based piece of equipment and is not a consumer product within the meaning of the Regulations and Specifications (GPSG).

The appliance must not be used without a thorough knowledge of the ”Installation and Operating Instructions“. Incorrect operation could result in injuries to personnel or damage to the appliance.

2.1 Operating panel

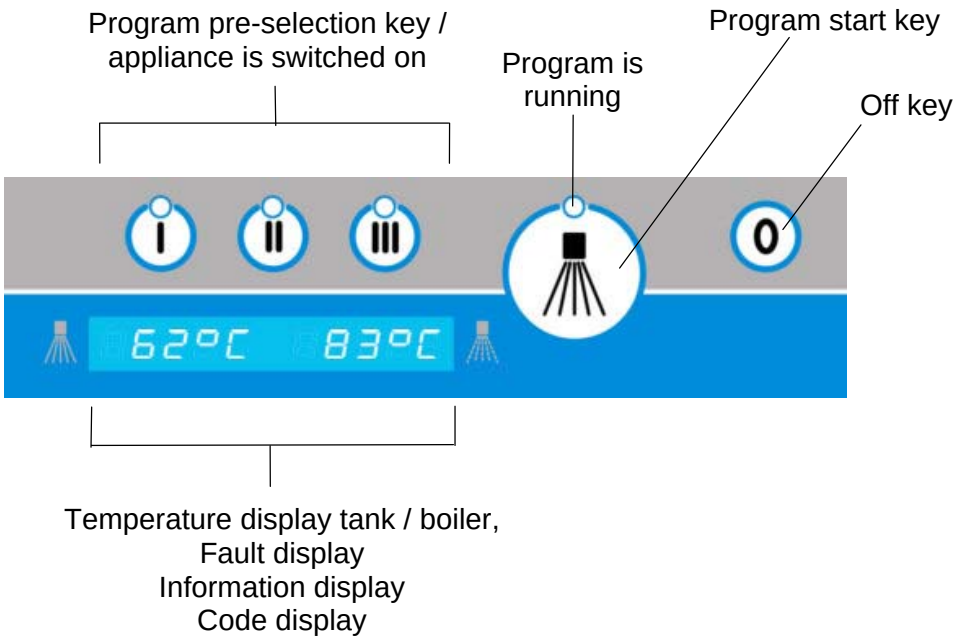


Illustration 1; Operating panel

Key / display	Meaning
	Short cycle for lightly soiled items – Wash program I
	Normal cycle – Wash program III
	Intensive cycle – Wash program IV
	Wash temperature
	Final rinse temperature
	Program start Tank drain Self-cleaning cycle
	Switch off appliance/ Cycle interruption

Table 1; Program key function / items to be washed

2.2 Preparation for washing and rinsing

The preparatory work described below must be carried out before each operation.

- Open the door
- Insert the sieves.
- Close the door.

Danger of crushing.
Close the appliance with both hands.

- Switch on the appliance by pressing one of the program pre-selection keys.

During the filling and heating phase, the light above the pre-selection key will flash. When the light remains constantly lit, the machine is ready for operation.
The time until the appliance is ready for operation depends on the temperature of the water supply and the heating capacity of the boiler or tank installed.
In the case of cold water supply, the time taken is:

FV 130.2 about 50 minutes.
FV 250.2 about 40 minutes.

2.3 Automatic dosing

The required detergent and rinse aid is moved from the containers into the tank or boiler via electronically controlled dosing units. The dosing is carried out automatically according to the requirements arising during the wash process.

2.4 Operation during washing and rinsing cycle

The following fundamental principles must be observed when placing the items to be washed in the baskets:

- All hollow containers must always to be loaded **upside down**. Otherwise the water will be trapped inside and they will not dry to a brilliant finish.
- Plates, trays and big plates should always stand **at a slight angle** in the basket with their inside faces pointing upwards.
- When using cutlery baskets, ensure that cutlery is always inserted handle down.
- Load the cutlery baskets with a **mixture** of spoons, knives and forks, as identical items of cutlery can be too close together.
- Do **not overload** the baskets.
- Do not stack the dishes in the wash basket directly on top of each other, as the wash water could not strike the items directly and unnecessarily long wash times would have to be selected. Short wash times with baskets which are not overloaded are much more economical.

2.4.1 Start the wash cycle

Program start key

- Place the items in the basket.
- Place the basket in the appliance, ensuring that it is correctly centred.
- Close the door.
- Press the program start key.

The appliance washes and rinses automatically and switches off the wash program after completion. The program cycle is indicated by a light on the program start key.

The wash time can differ from the set program time if the boiler heating capacity is not sufficient for heating up the fresh water to the pre-set boiler temperature during the program time. In this case, the automatic wash time extension is activated.

2.4.2 Remove the washed items

- When the light goes out, open the door and remove the basket.

3

Shutting down the appliance

Off -key

- Press the ”0“key (OFF key). The machine is switched off when all the lights are out.

Program start key

- Press the program start key to drain the tank.
- The tank interior is sprayed with clean hot water after the tank water has been drained. The door must remain closed. The waste water pump switches off automatically.

4

Care and maintenance

4.1 Care, general

The appliance has been designed to minimise the need for cleaning, care and maintenance.

However, correct care and maintenance is necessary for reliable, safe and long-lasting operation of the appliance and in the interest of hygiene and cleanliness.

To facilitate this procedure, a maintenance contract can be concluded with the manufacturer or the manufacturer’s agent.

Works/repairs which were not correctly executed and the use of unauthorised parts by unqualified personnel endanger both operators and the appliance, and will invalidate the warranty.

4.2 Refilling with detergent

The detergent container is situated close to the universal wash appliance.

- Check the level and, if necessary, replace the container by a full one.

Use only lather-free detergents suitable for industrial dish-washing machines.

4.3 Refilling with rinse aid

The rinse aid storage container is situated close to the universal automatic dishwasher.

- Check the level and, if necessary, replace the container by a full one.

Use only lather-free rinse aids suitable for industrial dish-washing machines.

4.4 Cleaning

After the tank has been drained, proceed as follows:

- Food residues sticking to the tank, tank heating element and sieves must be removed with a brush.
- Dismantle the wash arms and rinse them under running water.
- Wash nozzles must be cleaned daily.
- The cleanliness of final rinse nozzles must be checked weekly and if necessary clean under running water.

The inserts for the final rinse nozzles must be inserted with the prongs facing the water flow.

4.5 Maintenance of stainless steel surfaces

The appliance is made of high-quality stainless steel. Nevertheless, under certain conditions corrosion may appear.

- To maintain stainless steel surfaces permanently free from corrosion use only cleaning products which are properly.

Use only products which do not attack the material, build up a film or cause discoloration.

4.6 De-scaling

If the appliance is operated with hard water, scale can build up in the boiler and wash tank, making it necessary to de-scale the tank interior, the tank heating equipment, the boiler heating equipment, the wash system and the items to be washed.

Use only products suitable for industrial dishwashing machines to de-scale the appliance. When using these products please observe the manufacturer’s instructions.

After de-scaling:

- Flush the de-scaling product completely out of the appliance. 1 or 2 rinse cycles with fresh water are necessary to achieve this.

Even small amounts of de-scaling product remaining in the appliance may cause plastic components and sealing materials to deteriorate.

If the appliance is heavily scaled, you should ask a service engineer from the agency responsible to de-scale the boiler.

5 Machine with built-in AktivClean water softening device

5.1 General

The built-in ActivClean water softening device operates automatically. All that is necessary is to replenish the salt from time to time. The regeneration water is fed directly into the drain. It is not necessary to drain the tank during the regeneration process.

The temperature of the feed water must not exceed 60° C.

5.2 Adjustment of water hardness

The water softening device is pre-set to 16°Gh in the factory. When the service engineer installs or commissions the appliance he should adjust this value depending on the actual water hardness. Should there be any further changes in the water hardness, this parameter must be adjusted accordingly as set out in the Short Programming Instructions.

5.3 Capacity of the built-in water softening device between two regenerations

Hardness (°Gh)	Capacity (l)
8	250
10	200
12	167
14	143
16	125
18	111
20	100
22	91
24	83
26	77
28	71
30	67
32	63
34	59
36	56
38	53
40	50
42	48

5.4 Regeneration

The automatic regeneration is indicated by a yellow light. The regeneration lasts approx 8 minutes.

A further programme sequence can take place during this time. If a second programme cycle is started, the washing time is extended until the regeneration is completed. The draining pause and the fresh water final rinse only take place after the regeneration is complete.

A red light indicates when the salt container has to be re-filled. The salt container is located in the tank. It will take approximately 1.5 kg of regeneration salt, grain size 0.3 – 1 mm. This quantity is sufficient for about 12 regeneration cycles. Use a funnel for adding the salt to the container. The seal and the thread of the salt solution container must be cleaned before closing the container.

5.5 Fault indications

Red light is flashing:
The water softening device is exhausted and supplies only hard water.
➢ Re-fill with salt. You should be aware that the consequence of continuing to use the appliance with an exhausted water softening device could be loss of capacity or the machine could even become unserviceable.

6 Trouble-shooting

Despite being expertly designed, the machine may develop minor faults which are usually easy to eliminate. This section explains a number of possible problems and how you can deal with them yourself.

Before carrying out work on the appliance it must be disconnected from the power supply. The local main switch must be switched ”OFF“, or the local main fuse must be removed.

Should any of the described faults arise repeatedly, their cause must be established.

Faults not described here can in general only be eliminated by a technician or electrician. Please contact the agency responsible or the authorised dealer.

6.1 Information messages and trouble-shooting

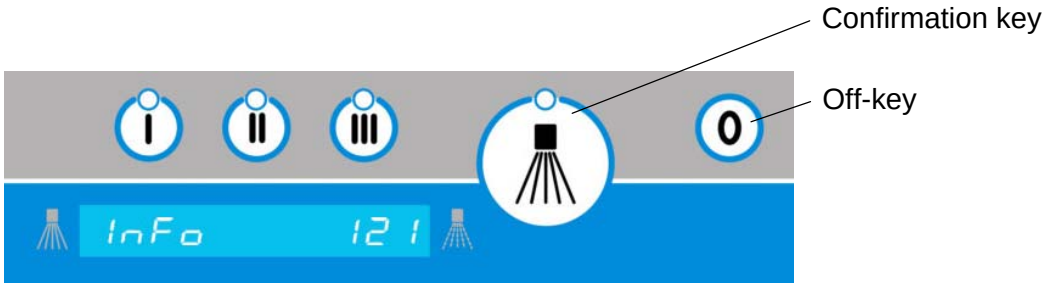


Illustration 2: Information display

Information displays can be deleted using the confirmation key.
Provided that the operation of the machine has been established, the next program sequence will begin.

Pressing the “OFF” key will also delete the information indication.

Information display (extract)

INFO-No.	Description	Possible cause
120	Emergency program active Restricted washing possible	No boiler / tank heating No fresh water supply Check system
121	Door not closed	Switch lever released Check connection S1 Change micro-switch Check adjustment micro-switch
122	Incorrect password / no authorization	Enter code again

Table 2: Information display

6.2 Error reports and troubleshooting

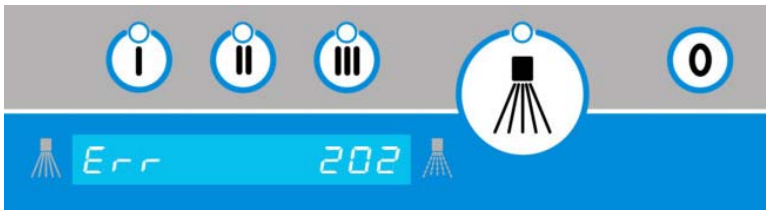


Illustration 3: Fault indication

Fault indications will be automatically deleted once the fault has been corrected.

Fault indications (extract)

ERR.-No.	Description	Possible cause
111	Floor pan leakage	Leakage inside the machine Pump sump / motor / etc. Defective leaking water switch Repair fault, remove water
112	Dosage leakage	Dosage pump leakage Hose defective / kinked Defective dosage pumps Defective measurement electrodes

201	Level not reached during first filling.	Fresh water supply insufficient (water faucet closed) Aquastop hose kinked Inlet strainer soiled Aquastop defective Boiler switch defective
202	Level not reached at right time during filling	See 201
301	Number of pumping cycles exceeded Tank level analysis disrupted	Booster pump discharge too low Rinse jets soiled Air trap soiled Booster pump rotor defective Condensate in level pipe Hose kinked / loose / not watertight
302	When pumping out during rinse program, level does not fall below level 1.	Wash water pump yield too low Wash water pump soiled / defective Rotor loose Wash water pump plug connector loose Start capacitor defective Tank level analysis disrupted Aquastop not closing completely No signal from ON/OFF circuit boards
303	Level does not fall below level 3 after time (LP ON)	See 302
306	Maximum level value exceeded Tank level analysis disrupted	Ventilation valve soiled Check tank level Check level sensor air trap / hose

Table 3: Fault displays

Please notify a customer service technician if information or fault numbers not shown in the tables appear, or if the suggested measures do not eliminate the fault.

7 Appendix

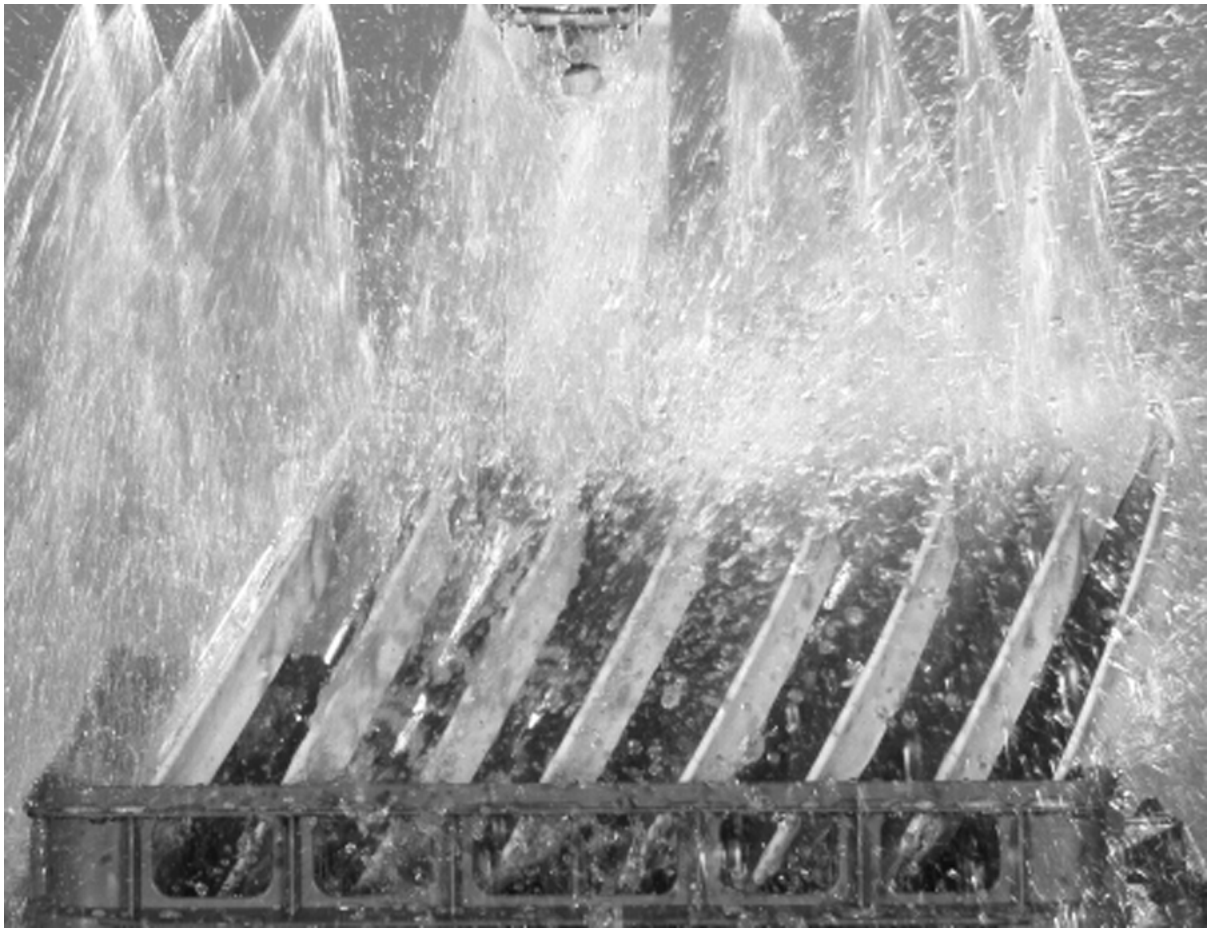
7.1 Name and address of manufacturer

MEIKO Maschinenbau GmbH & Co. KG
Englerstrasse 3
D-77652 Offenburg
Telephone: 0781 / 203 - 0
Fax: 0781 / 203 - 1179
www.meiko.de
e-mail: info@meiko.de

Operating instructions

Universal warewashing machine
FV 130.2
FV 250.2

Service instructions



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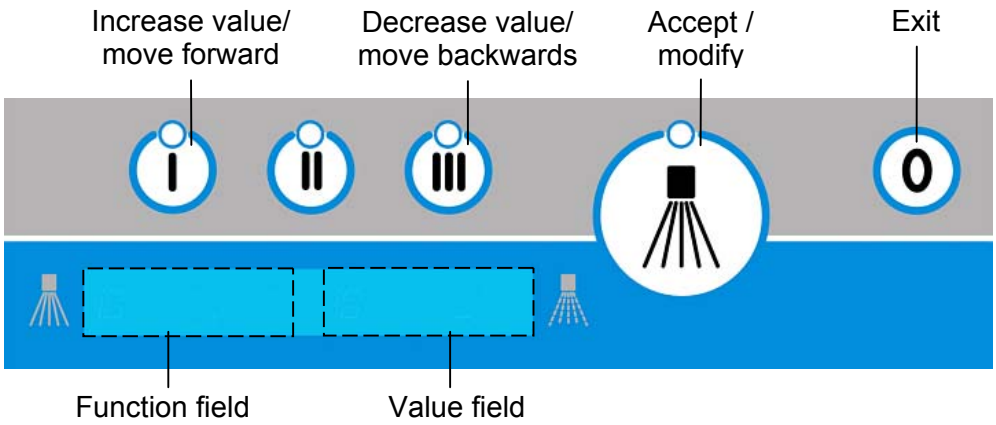
2 Authorized user of this documentation

The works described in this booklet may only be carried out by specialists of the manufacturer, the responsible agency or an authorized dealer.

3 Settings / modifications / on-site adaptation

3.1 Using the keyboard for programming



Access codes for various user-levels have been defined. Once the complete code has been entered, the entered code is compared with the internal code table. Depending on the code entered, the corresponding user level will be accessed. 2 access codes are available for each user level; the first is for restricted access, i.e. no modification of parameters is possible (viewing mode), and the second gives access to the entire range of functions (viewing and modification).

In the short programming instructions accompanying every machine in the series, this is described in condensed form.

For control programming, the power supply must be available but the machine must be completely switched off (no LED must be illuminated).

Code – input:

View service data:	CODE 10000
Modify service data:	CODE 10001
View configuration data:	CODE 20000
View dosing technology:	CODE 40000
Modify dosing technology:	CODE 40044

The code numbers for the further levels can be found in the Service Manual.

3.2 Code entry

To get into the code entry mode, you should keep the key “0” pressed (for around 3 seconds) until you see

on the display unit.

By pressing the key “0” once again you can leave the programming area at any time.

The digit to be modified will flash.

Press the “I” key to increase the value/code indicated on the display unit, or press the “III” key to decrease it, or press the “accept” key to save it. The next value will then flash and will be the only one visible.

If your entry is incorrect you will exit the code entry procedure, and the information code 122 will be displayed.

If you enter all the digits correctly you will arrive at the chosen level, either service, configuration or machine data.

3.3 Service level

The list of service parameters can be found on this level (parameter numbers 1xx). Here you can view these or modify them, or you can also call up the ventilation of the rinse and wash hoses.

On the service level, you will first see the display below:

This corresponds to viewing / modifying parameters. (see 3.3.1)

This corresponds to rinse aid inlet ventilation. (see 3.3.2)

This corresponds to detergent inlet ventilation. (see 3.3.3)

This corresponds to manual start to regeneration (see 3.3.4)

Press the “I” key to move forwards, press the “III” key to move backwards, or press the key “accept” to make a selection. You are now at the current level.

You can leave this level by pressing the “0” key.

3.3.1 View/modify parameters

Confirm

the indication by pressing the “accept” key.

Now, the first parameter will be displayed with a value.

Press the “I” key to go forwards and the “III” key to go backwards, until the parameter you require is displayed.

Press the “accept” key to confirm the parameter modification, the value will flash. Use the “I” key to increase the value, or the “III” key to decrease the value, and the “accept” key to save.

You can leave this level by pressing the “0” key.

See 3.6 for the parameter list

3.3.2 Ventilating the rinse aid inlet

Confirm by pressing the “accept” key.

Now the dosage pump will be activated and the remaining running time will be indicated.

By pressing the “0” key, you can leave this level. The ventilation will be interrupted.

3.3.3 Ventilating the detergent inlet

Confirm by pressing the “accept” key.

Now the dosage pump will be activated and the remaining running time will be indicated.

You can leave this level by pressing the “0” key. The ventilation will be interrupted off.

Should the ventilation process be insufficient, repeat the process.

3.3.4 Manual start to regeneration

Confirm by pressing the “accept” key.

The regeneration is started now and the remaining cycle time is indicated.

This level can be left by pressing the "0" key. The regeneration is interrupted. The regeneration re-commences when the machine is switched on again.

3.4 Configuration level

You can find the list of configuration parameters on this level (parameter numbers 2xx). Here you can view these and modify them. You can also call up the status of the inputs and outputs, or set the outputs for testing.

On the service level, you will first see the display below:

This corresponds to the viewing/modifying parameters. (see 3.4.1)

This corresponds to viewing the status of inputs. (see 3.4.2)

This corresponds to viewing and setting the status of outputs (see 3.4.3).

Press the “I” key to move forwards or the “III” key to move backwards or the “accept” key to make a selection. You are now at the current level.

Press the “0” key to leave this level.

3.4.1 Viewing / modifying parameters: (depending on the code entered)

Confirm

this display by pressing the “accept” key.

Now, the first parameter will be displayed with a value.

Press the “I” key to move forwards or press the “III” key to move backwards, until the parameter you require is displayed.

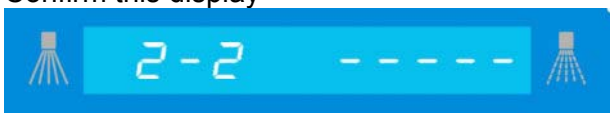
Confirm the parameter to be modified by pressing the “accept” key, the value will flash. Press the “I” key to increase the value, the “III” key to decrease the value, and the “accept” to save the value.

You can leave this level by pressing the “0” key.

See 3.6 for list of parameters.

3.4.2 Viewing input status:

Confirm this display



by pressing the “accept” key.

Now, the first input will be displayed, with the status

.

Press the “I” key to move forwards and the “III” key to move backwards, until you reach the input you require.

Display: input set

Display: input not set

By pressing the “0” key, you can leave this level.

Assignment details for the inputs are given on the assignment list for each machine (see 3.7).

3.4.3 Viewing / modifying output status (according to code entered)

Confirm

this display by pressing the “accept” key.

Viewing:

Now, the first output will be shown, with status.

.

Press the “I” key to move forwards or press the “III” key to move backwards, until the output you require is displayed.

Modifying:

Press the “accept” key to confirm the modification of the output, the value will flash. Press the “I” key to modify the value and press the “accept” key to save it.

The output is now set.

You can leave this level by pressing the “0” key.

Assignment details for the outputs are given on the assignment list for each machine (see 3.7).

3.5 Viewing / modifying dosing technology level

By entering code 40000 (read only) or 40044 (read / enter), the user can access the new 4th parameter level summarizing all the dosing technology parameters:

P104, P105, P218, P219, P224, P225, P321, P322, P326, P327.

See 3.6 for list of parameters.

3.6 Parameter list

Par. No.	Configuration options	Use as	Value range	Unit	Factory setting	Observation
101	Rinse program key 1	Parameter	1 .. 50	-	1	Assign rinse program no. to key 1; assignment adjustable
102	Rinse program key 2	Parameter	1 .. 50	-	3	Assign rinse program no. to key 2; assignment adjustable
103	Rinse program key 3	Parameter	1 .. 50	-	4	Assign rinse program no. to key 3; assignment adjustable
104	Rinse aid dosage	Parameter	0.10 .. 1.00	ml/liter water	0.2	Value can be read from the rinse aid container label (dependant on water quality)
105	Detergent dosage	Parameter	0.1... 20.0	ml/liter water	2.0	Value can be read from the detergent container label (dependant on water quality)
107	Beep ON/OFF	Parameter	0/1	-	1	Switch on/off acoustic ready message
111	Total running time indication	Indication	5 digits	hrs.		Running time, query only
112	Total number of wash cycles	Indication	5 digits	-		Wash cycles/loads, query only
113	Total number of wash cycles since last reset	Indication	5 digits	-		Wash cycles/loads, re-setting possible.
114	Serial number	Indication	8 digits	-		Factory settings can be called up
119	IR-communication	Parameter	0/1	-	1	It is possible to shut off communication via IR interfaces.
120	Factory setting load service parameter	Parameter	0/1	-	0	Effective only upon power supply reset ON/OFF. **Important**! All changes to service parameters will be reversed. Power supply reset must be carried out within 5 minutes, otherwise factory settings will not be loaded. Without power supply reset, the information 123 will be displayed.

Par. No.	Configuration options	Use as	Value range	Unit	Factory setting	Observation
201	Machine model	Parameter	1 ... 4	-		1: FV40.2 2: FV130.2 / FV 250.2 3: DV80.2 4: DV120.2 **Important!** Only assignment list and machine sequences are changed – no parameters changed
202	Tank target temperature	Parameter	10 ... 80 (50 .. 176)	°C/°F	60	Standard for all the rinse programs on one appliance! Output dependent on definition.
203	Pre-rinse time	Parameter	0 ... 8	Sec.	0	See pre-rinse process step
204	Post-rinse time	Parameter	4 ... 30	Sec.	7	Time the booster pump activated (running time limited by P306)
205	Operating status display	Parameter	0 ... 8		1	Definition of the information which is to be switched via the potential-free contact (see KD), e.g.: 0 – no information 1 – F/H, ready for washing/washing or pumping out 2 – F/H, washing or ready for washing 3 – F/H (Filing/ Heating) 4 – Ready for washing 5 – Washing 6 – Pumping out 7 – Error 8 – Not status machine OFF and Draining 9 – reserve 10- not status Machine OFF
218	Shortage of rinse aid	Parameter	0/1		0	Monitoring Display
219	Shortage of detergent	Parameter	0/1		0	Monitoring Display
224	Rinse aid pump activation mode	Parameter	0 .. 4	-	1	Definition: Activation of rinse aid pump: 0 – rinse pump = 0; no signal 1 – activate rinse aid pump according to calculated running time

Par. No.	Configuration options	Use as	Value range	Unit	Factory setting	Observation
						2 – rinse aid pump = booster pump; activated as booster pump 3 – rinse aid pump = wash pump; activate as wash pump 4 - free
225	Detergent pump activation mode	Parameter	0 .. 4		1	Definition: Activation detergent pump: 0 – Detergent pump; no signal 1 – Detergent pump; activate according to calculated running time 2 – Detergent pump = booster pump; activate as booster pump 3 – Detergent pump = wash pump; activate as wash pump 4 – Option – detergent pump using negative pressure dosing.
240	Load factory settings for configuration data	Parameter	0/1	-	0	Effective only upon power supply reset ON/OFF. **Important**! All changes to service parameters will be reversed. Power supply reset must be carried out within 5 minutes otherwise factory settings will not be loaded. Without power supply reset, the information 123 will be displayed.
321	Rinse agent pump output	Parameter	0.1 ...10	l/h		Rinse agent pump. Output definition.
322	Detergent pump output	Parameter	0.1 ...20	l/h		Detergent pump. Output definition.
326	Rinse agent pipe vent time	Parameter	0 ... 255	sec		Activate rinse agent pump temporarily to remove air from pipe.
327	Detergent pipe vent time	Parameter	0 ... 100	sec		Activate detergent pump temporarily to remove air from pipe.

3.7 Assignment list
View inputs / control outputs

Indication		Input / output / other	Conditions
Left	Right		
In 1	0/1	Door closed	none
In 2	0/1	Boiler level	none
In 3	0/1	Leak water switch floor	none
In 4	0/1	not occupied	none
In 5	0/1	not occupied	none
In 6	0/1	not occupied	none
In 7	0/1	not occupied	none
In 8	0/1	not occupied	none
In 9	0/1	not occupied	none
In 10	0/1	not occupied	none
In 11	0/1	Leak water switch dosage	none
In 12	0/1	not occupied	none
In 13	0/1	Threshold tank level 1	none
In 14	0/1	Threshold tank level 2	none
In 15	0/1	Threshold tank level 3	none
In 16	0/1	Tank level 4 (option)	none
In 17	0 .. 255	Without function	none
In 18	0 .. 255	Without function	none
In 19	xxx	Boiler temperature in °C or °F	none
In 20	xxx	Tank temperature in °C or °F	none
In 21	xxx	Tank level (1 mm unit)	none
In 22	0 .. 255	Without function	none
Ou 1	0/1	Booster pump	No leak water
Ou 2	0/1	Swivel wash arm	No leak water
Ou 3	0/1	Drain pump	No leak water
Ou 4	0/1	Rinse aid – dosage pump	No leak water
Ou 5	0/1	Detergent – dosage pump	No leak water
Ou 6	0/1	Operation indicator	No leak water
Ou 7	0/1	Filling valve	No leak water
Ou 8	0/1	SASm soft starter system	No leak water
Ou 9	0/1	Boiler heating	No leak water
Ou 10	0/1	Tank heating	No leak water; no boiler heating active
Ou 11	0/1	Wash pump	No leak water
Ou 12	0/1	Pressure reduction (option)	No leak water

Leak water switch condition: Leak water switch must not have operated.
Heating condition: Tank / boiler heating are inter-locked (boiler priority)
Tank heating only when boiler heating deactivated.

3.8 Rinse program parameter update: 11.06.2002

Rinse Program no.:	Boiler temperature target value	Wash time target value		Wash pressure reduction target value
		Washing	Total	
1	83	95	120	0
2	83	155	180	0
3	83	215	240	0
4	83	335	360	0
5	83	455	480	0
6	65	95	120	0
7	65	155	180	0
8	65	215	240	0
9	65	335	360	0
10	65	455	480	0
11	83	95	120	1
12	83	155	180	1
13	83	215	240	1
14	83	335	360	1
15	83	455	480	1
16	65	95	120	1
17	65	155	180	1
18	65	215	240	1
19	65	335	360	1
20	65	455	480	1
21				

The dosage times will be adapted to the rinse time, so that the correct concentration remains if the rinse time is modified.

3.9 Information reporting and troubleshooting

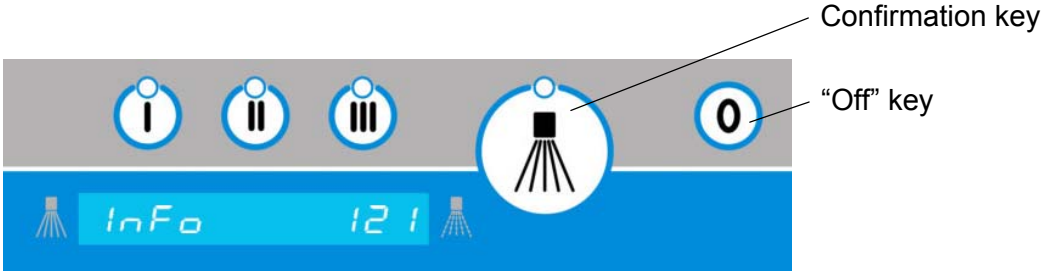


Figure 2: Information display

Information displays can be cleared by pressing the confirmation key.

Provided that the machine function is restored, the next program sequence will begin.
The information display can also be deleted by pressing the “Off” key.

Information indicator (extract)

INFO-no.	Description	Possible cause
120	Emergency program active Restricted washing possible.	No boiler / tank heating No fresh water supply Check system
121	Door not closed	Switch lever loosened Check connection S1 Change microswitch Check microswitch adjustment Replacing a defective I/O circuit board
122	Incorrect password / no authorization	Enter code once again
123	Factory setting parameter list	Switch power supply ON/OFF within 5 minutes and re-set to factory settings. This will be rejected and parameters will be retained. Information 123 will disappear.

Table 2: Information displays

3.10 Error messages and troubleshooting

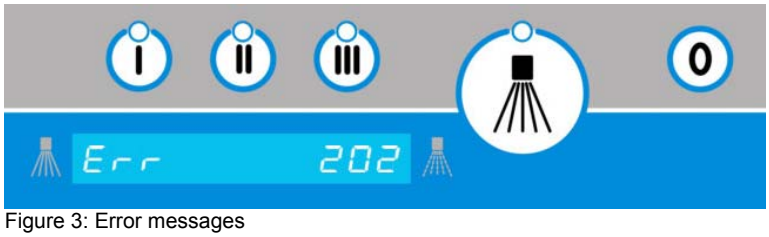


Figure 3: Error messages

Error messages will disappear automatically when the fault has been rectified.

Error messages (extract)

ERR.-no.	Description	Possible cause
001	EEPROM plug-in fault.	EEPROM not available / incorrectly plugged in / defective Empty or incorrect EEPROM Replace EEPROM with correct parameter set
111	Floor pan leakage	Leak inside the machine Pump sump / motor /etc. Defective leak water switch Repair fault, remove water
112	Dosage leakage	Dosage pump leak Hose defective / kinked Defective dosage pump Defective measurement electrodes
201	Level not reached during 1st filling	Fresh water supply insufficient (water faucet closed) Aquastop hose kinked Inlet filter soiled Aquastop defective Boiler switch defective
202	Level not reached early enough during filling	See 201
203	No change detected by the level switch during emptying	Boost pump defective Pressure increasing pump plug connector loosened Start capacitor defective Plug connector loosened Boiler level switch defective No boost pump signal to - from input/output circuit boards Check boost pump / S2 using manual control
204	Following the end of rinse time, still no change detected by the level switch	See 203

ERR.-Nr.	Description	Possible cause
205	Temperature increase not achieved	Boiler heating defective / thermal fuse Radiator Temperature sensor defective, incorrect installation position Boiler contactor defective, performance switch loose No signal from input/output circuit boards
206	Wash time increase	Boiler not ready for rinsing early enough (boiler level/ boiler temperature) Boiler heating defective / thermal fuse, radiator Temperature sensor defective Boiler contactor defective, performance switch loosened No signal from input/output circuit boards
210	Temperature sensor short circuit	Check sensor cable (plug contacts) Replace sensor Install sensor correctly
211	Temperature sensor interruption	See 210
212	Actual boiler temperature too high	Contactor sticking Incorrect sensor / defective sensor Check sensor / cable (contact plug MIKE II XA5)

ERR.-No.	Description	Possible cause
301	Number of circulatory pumping cycles exceeded. Tank level value disrupted.	DSP yield too low Rinse jets soiled Air trap soiled Booster pump rotor defective Condensate in level pipe Hose kinked / loose / not watertight
302	When pumping out during rinse program, level does not fall below level 1.	Drain pump yield too low Drain pump soiled / defective Rotor loose Drain pump plug connector loose Start capacitor defective Tank level analysis disrupted Aquastop not closing completely No signal from ON/OFF circuit boards
303	Level does not fall below level 3 after time (drain pump ON)	See 302
304	Temperature increase not reached	Boiler heating defective / thermal fuse radiator Temperature sensor defective, incorrect installation position Tank protection defective, performance switch loose
305	Boiler content quantity insufficient for rinsing. Level 2 not reached	See 301 Level switch defective Plug connector loosened
306	Max. level value exceeded Tank level analysis disrupted	Ventilation valve soiled Check tank level Level sensor air catch / check hose
307	Tank level sensor defective	Connection plug loosened Sensor defective Replace input/output circuit boards
310	See 210	See 210
311	See 211	See 211
312	See 212	See 212

Table 3: Error messages

Should information or fault numbers not shown in the tables be indicated, or should the suggested measure not lead to the elimination of the fault, please notify a customer service technician.

4 Maintenance

4.1 Dosing units

The dosing units themselves are basically maintenance-free.

4.1.1 Change of products

Change of product means that one rinse aid or detergent product is replaced by another. The use of differing products alongside each other can result in break-downs.

- Hose lines and dosing units must always be rinsed out with warm water.

4.2 De-scaling

If the appliance was operated with hard water, the boiler and wash tank could have lime scale deposits. De-scaling of the tank interior, boiler housing, tank heating, boiler heating and wash and final rinse system then becomes necessary.

For de-scaling the appliance use only products suitable for industrial dishwashers. Please observe the instructions of the manufacturers of such products.

After de-scaling the appliance:

- Remove the de-scaling agent completely from the appliance. 1 or 2 rinse cycles with fresh water are necessary to achieve this.

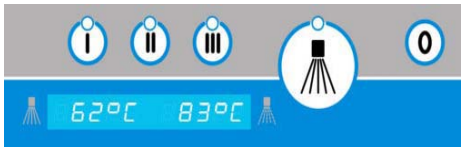
Even small residues of de-scaling agents can be sufficient to destroy plastic parts and sealing materials!

5 Appendix

5.1 Name and address of manufacturer

MEIKO Maschinenbau GmbH & Co. KG
Englerstrasse 3
D-77652 Offenburg
Phone no.: (+ 49) 781 / 203 - 0
Fax no.: (+ 49) 781 / 203 - 1121
www.meiko.de
e-mail: info@meiko.de

SHORT OPERATING INSTRUCTIONS
FV 130.2 / FV 250.2



Control panel

1. Preparing to wash and rinse

- Open the door.
- Insert suction sieve and tank covering sieve.
- Close the door.
- Turn on the machine by pressing one of the pre-selector buttons.
- Check level and if necessary refill detergent and rinse aid reservoirs.
- The machine is ready for operation when flash light of pre-selector button stops.

2. Washing and rinsing

- Place the wash-ware to be washed in the basket.
- Insert the basket in the machine.
- Close the door.
- Program I for lightly soiled wash-ware.
- Program II for normal soiled wash-ware.
- Program III for heavily soiled wash-ware.
- Press the program start button.
- The machine automatically washes, rinses and switches itself off when the wash program is finished. Open the door after the lamp goes out and remove the basket.

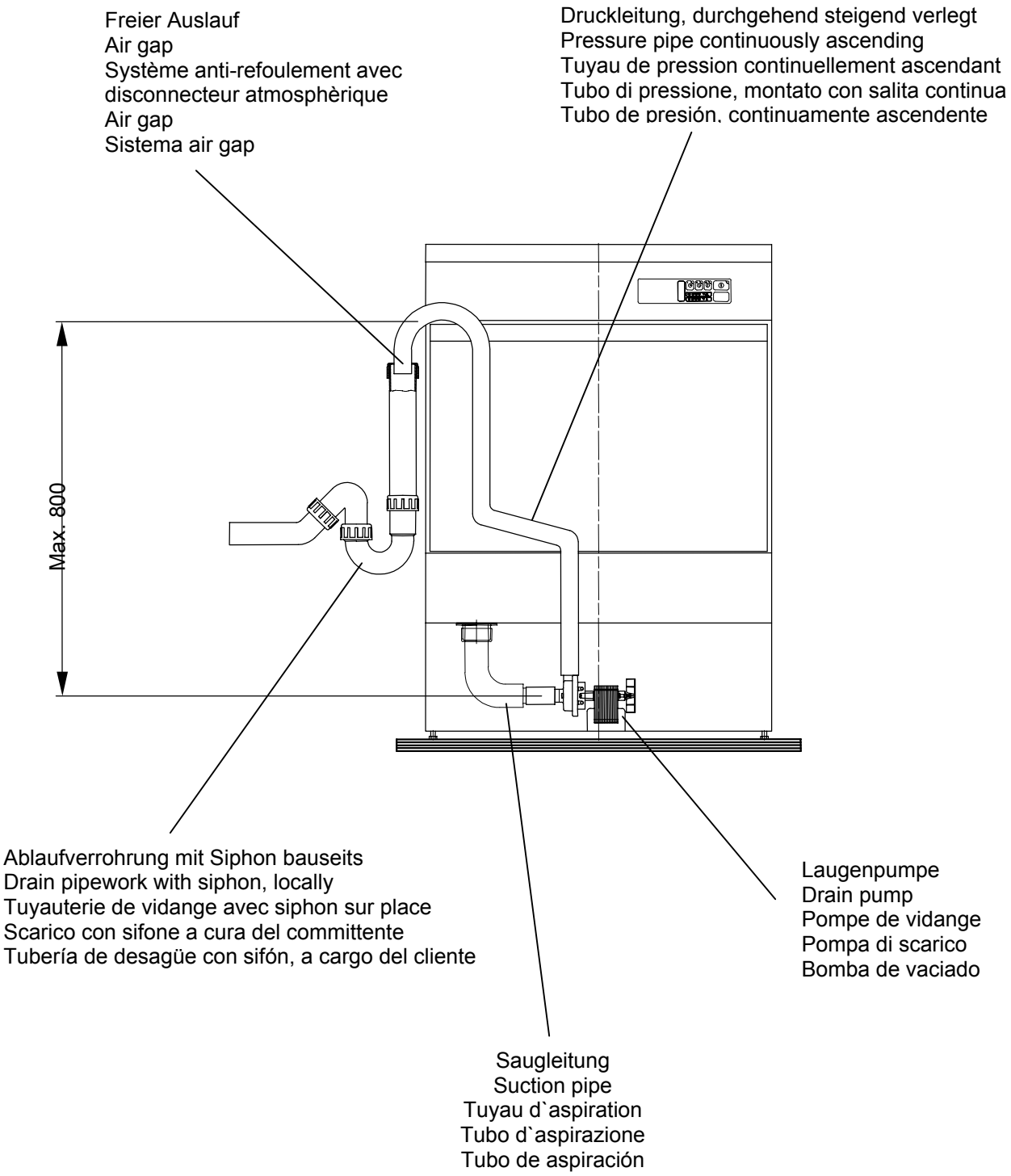
3. Shutting the machine down

- Press the “0“ button (Off button). The machine is switched off when all the lights are out.
- To empty the tank press the program start button.
- After the tank water has been pumped out, the interior is rinsed with fresh hot water. The door must be kept closed. The waste water pump disconnects itself automatically.
- Clean tank, strainers and wash arms.

Anschlußvorschrift für Laugenpumpe
Connection prescription for drain pump
Prescription de connexion pour pompe de vidange
Prescrizioni di collegamento per la pompa scarico
Prescripciones para la conexión de la bomba de vaciado

MEIKO

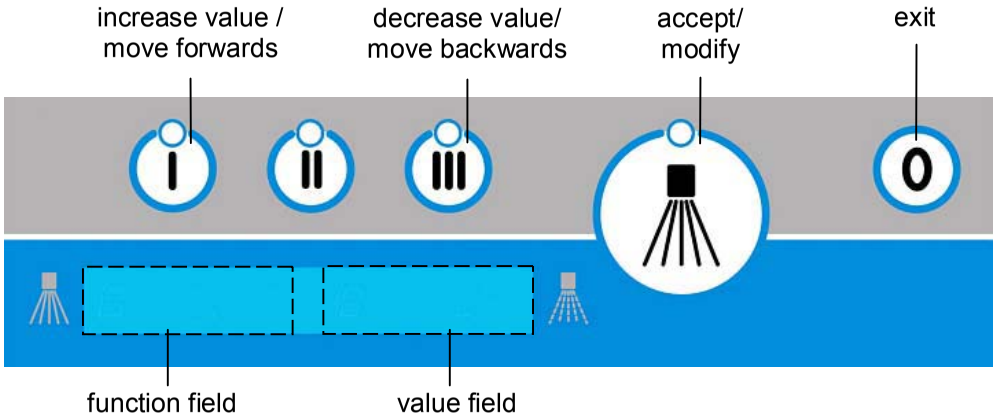
8.97/LP 5-sprachig.DOC



Quick programming instructions
FV 130.2

06.03/KE-FV130_2-MIKEII

Keyboard use during programming



General:
In order to program the controls, power must be available, but the machine must be completely switched off (no LED must be illuminated).

Code – entry:
View service data: **CODE 10000** Modify service data: **CODE 10001**

To access the code entry mode, you should keep the “0” key pressed (for around 3 seconds) until you see **Code 1----** on the display unit. You can leave the programming area at any time by pressing the “0” key once again,.
The digit to be modified will flash. Use the "I" key to increase the values in the display unit; these are then accepted by using the "accept" key. You should select each of the digits in this way, one after the other, until the code has been completely entered.
If your entry is incorrect, you will leave the code entry procedure, and information code 122 will be displayed. If you enter all the digits correctly, you will be given access to the service level.
You can move forwards using the “I” key, and backwards using the “III” key.

The following main items are available on the service level:

1-1 ----- View / modify parameters
1-2 ----- Ventilate rinse aid inlet
1-3 ----- Ventilate detergent inlet
1-4 ----- Manual start to regeneration

View parameters:

1-1 ----- Accept the item by using the “accept” key, or use the “I” or “III” keys to select the parameters. On the left, the parameter Pxxx will be displayed, and on the right, the value belonging to it.
You can use the “0” key to go back one level.

Modify parameters:

1-1 ---- Select the parameters in the same way as when viewing, but access is via the code for “modify service data”.
To modify one of the parameters, select it by using the “accept” key – the figure will flash.
Select the figure by using the “I” key or the “III” key, and save it by using the “accept” key.
You can leave the current level and go back to the previous level by using the “0” key.

Ventilate rinse aid inlet **1-2 -----**
Ventilate detergent inlet **1-3 -----**

Select the item you require. The dosing pump selected will be energized and the remaining running time will be displayed.
Exit the ventilation programme by using the “exit” key.

Manual start to regeneration:

The regeneration is started now and the remaining cycle time is indicated. This level can be left by pressing the "0" key. The regeneration is interrupted. The regeneration re-commences when the machine is switched on again.

Information/ error display:

If you receive an information message, you must follow the instructions. The rinse operation will still be possible.

If you receive an error message, no further rinse operation is usually possible.
Notify Customer Service.

Parameter mode

Important: the modification of factory set parameters may restrict the overall operation of the machine. In the event of arbitrary modification of parameters by unauthorized personnel, guarantee claims will become invalid.

Par. no.	Service parameters/ setting options	Use as	Value range	Unit	Factory setting	Observation
101	Rinse programme / key 1	Parameter	1 .. 50	-	1	Assign rinse program no. to key 1; this may be changed
102	Rinse programme / key 2	Parameter	1 .. 50	-	3	Assign rinse program no. to key 2; this may be changed
103	Rinse programme / key 3	Parameter	1 .. 50	-	4	Assign rinse program no. to key 3; this may be changed
104	Rinse aid dosing quantity	Parameter	0.10 .. 1.00	ml/liters of water	0.2	Value must first be set in outline according to manufacturer's information, but then it must be adjusted depending on the water hardness and the rinse result.
105	Detergent dosing quantity	Parameter	0.1... 20.0	ml/liters of water	2.0	Value must be set in outline according to manufacturer's information, but then it must be adjusted depending on the water hardness and the rinse result.
106	Hardness degree	Parameter	0 .. 50	°dH	30	The quantity of soft water available between two regenerations depends on the hardness of the water.
107	Switch beep on / off	Parameter	0/1	-	1	Switch acoustic " beep" ready message on / off
111	Total operating time display	Display	5-digit	Hrs.	0	Operating time, query only
112	Total number of wash cycles	Display	5-digit	-	0	wash cycles / loads, query only
113	Number of wash cycles since last reset	Display	5-digit	-	0	wash cycles / loads, resetting possible
114	Serial number	Display	8-digit	-	Serial number	Factory settings may be queried Important: the first 5 digits will be displayed in rotation with the last 3 digits!
119	IR-communication	Parameter	0/1	-	1	It is possible to shut off communication via IR interfaces.
120	Load factory setting service parameters	Parameter	0/1	-	0	Effective only upon power supply reset ON / OFF. **Important**! All changes to service parameters will be reversed. Power supply reset must be carried out within 5 minutes, otherwise factory settings will not be loaded. Without power supply reset, the information 123 will be displayed.

Rinse programme parameters 101-103

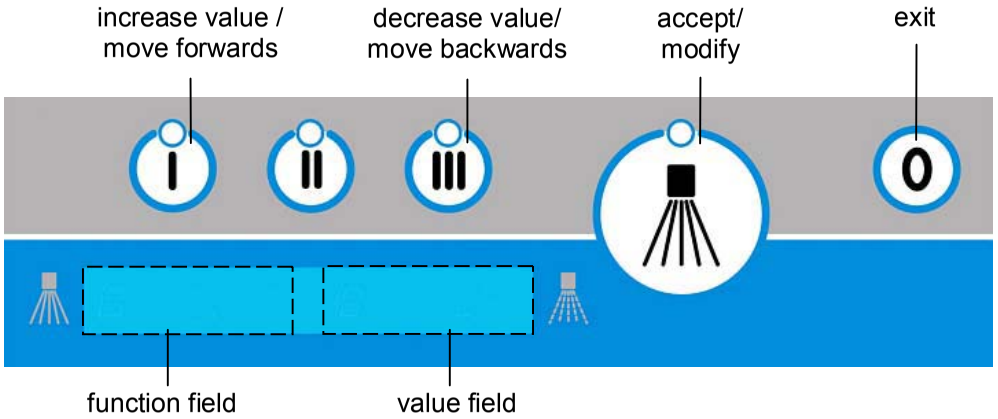
Rinse programme no.:	Boiler temperature target value	Wash time target value		Wash pressure reduction target value
		Washing	Total	
1	83	95	120	0
3	83	215	240	0
4	83	335	360	0

Details of further pre-defined rinse programmes are given in the Service Instructions.

Quick programming instructions
FV 250.2

06.03/KE-FV250_2-MIKEII

Keyboard use during programming



General:
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Code – entry:
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To access the code entry mode, you should keep the “0” key pressed (for around 3 seconds) until you see **Code 1----** on the display unit. You can leave the programming area at any time by pressing the “0” key once again,.
The digit to be modified will flash. Use the "I" key to increase the values in the display unit; these are then accepted by using the "accept" key. You should select each of the digits in this way, one after the other, until the code has been completely entered.
If your entry is incorrect, you will leave the code entry procedure, and information code 122 will be displayed. If you enter all the digits correctly, you will be given access to the service level.
You can move forwards using the “I” key, and backwards using the “III” key.

The following main items are available on the service level:

1-1 ----- View / modify parameters
1-2 ----- Ventilate rinse aid inlet
1-3 ----- Ventilate detergent inlet
1-4 ----- Manual start to regeneration

View parameters:

1-1 ----- Accept the item by using the “accept” key, or use the “I” or “III” keys to select the parameters. On the left, the parameter Pxxx will be displayed, and on the right, the value belonging to it.
You can use the “0” key to go back one level.

Modify parameters:

1-1 ---- Select the parameters in the same way as when viewing, but access is via the code for “modify service data”.
To modify one of the parameters, select it by using the “accept” key – the figure will flash.
Select the figure by using the “I” key or the “III” key, and save it by using the “accept” key.
You can leave the current level and go back to the previous level by using the “0” key.

Ventilate rinse aid inlet **1-2 -----**
Ventilate detergent inlet **1-3 -----**

Select the item you require. The dosing pump selected will be energized and the remaining running time will be displayed.
Exit the ventilation programme by using the “exit” key.

Manual start to regeneration:

The regeneration is started now and the remaining cycle time is indicated. This level can be left by pressing the "0" key. The regeneration is interrupted. The regeneration re-commences when the machine is switched on again.

Information/ error display:

If you receive an information message, you must follow the instructions. The rinse operation will still be possible.

If you receive an error message, no further rinse operation is usually possible.
Notify Customer Service.

Parameter mode

Important: the modification of factory set parameters may restrict the overall operation of the machine. In the event of arbitrary modification of parameters by unauthorized personnel, guarantee claims will become invalid.

Par. no.	Service parameters/ setting options	Use as	Value range	Unit	Factory setting	Observation
101	Rinse programme / key 1	Parameter	1 .. 50	-	1	Assign rinse program no. to key 1; this may be changed
102	Rinse programme / key 2	Parameter	1 .. 50	-	3	Assign rinse program no. to key 2; this may be changed
103	Rinse programme / key 3	Parameter	1 .. 50	-	5	Assign rinse program no. to key 3; this may be changed
104	Rinse aid dosing quantity	Parameter	0.10 .. 1.00	ml/liters of water	0.2	Value must first be set in outline according to manufacturer's information, but then it must be adjusted depending on the water hardness and the rinse result.
105	Detergent dosing quantity	Parameter	0.1... 20.0	ml/liters of water	2.0	Value must be set in outline according to manufacturer's information, but then it must be adjusted depending on the water hardness and the rinse result.
106	Hardness degree	Parameter	0 .. 50	°dH	30	The quantity of soft water available between two regenerations depends on the hardness of the water.
107	Switch beep on / off	Parameter	0/1	-	1	Switch acoustic " beep" ready message on / off
111	Total operating time display	Display	5-digit	Hrs.	0	Operating time, query only
112	Total number of wash cycles	Display	5-digit	-	0	wash cycles / loads, query only
113	Number of wash cycles since last reset	Display	5-digit	-	0	wash cycles / loads, resetting possible
114	Serial number	Display	8-digit	-	Serial number	Factory settings may be queried Important: the first 5 digits will be displayed in rotation with the last 3 digits!
119	IR-communication	Parameter	0/1	-	1	It is possible to shut off communication via IR interfaces.
120	Load factory setting service parameters	Parameter	0/1	-	0	Effective only upon power supply reset ON / OFF. **Important**! All changes to service parameters will be reversed. Power supply reset must be carried out within 5 minutes, otherwise factory settings will not be loaded. Without power supply reset, the information 123 will be displayed.

Rinse programme parameters 101-103

Rinse programme no.:	Boiler temperature target value	Wash time target value		Wash pressure reduction target value
		Washing	Total	
1	83	95	120	0
3	83	215	240	0
5	83	455	480	0

Details of further pre-defined rinse programmes are given in the Service Instructions.

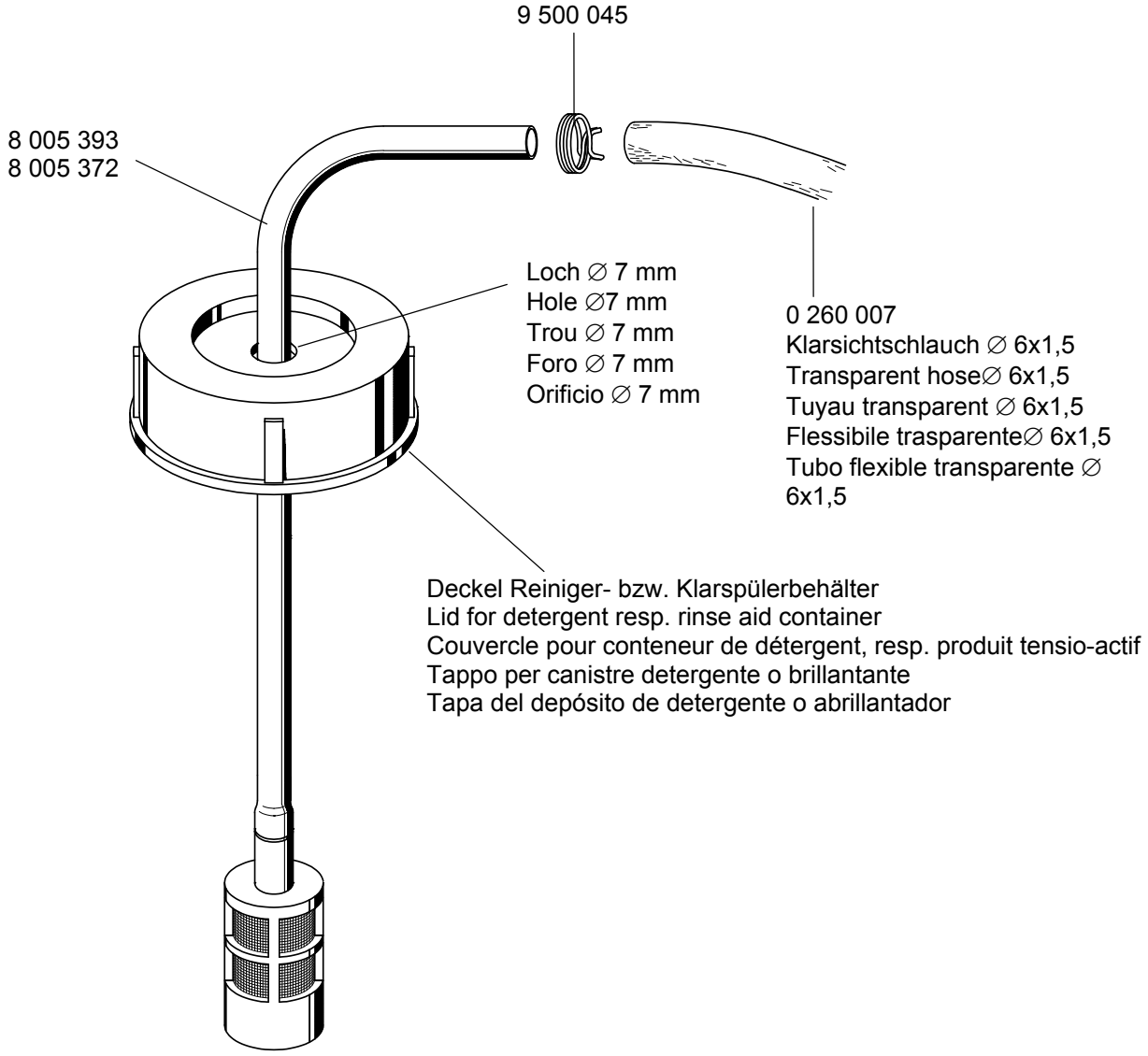
Service plan for stationary rack machines

Service procedures	FV 28G / FV28GIO EcoStar 430 F / 530 F EcoStar 530 F-M	FV 40.2 / FV 60.2 / FV 70.2 D	GK 60	OR 50 H	DV 40 – DV 240 B EcoStar 545D / DV 80.2 / DV 120.2 / DV 200.2 / DV 200.2 PW	DV 270 B	FV 130.2 – FV 250.2 DV 270.2	Component OK	Component faulty	Component replaced
1. Pumps										
Check pumps for watertightness, pump rotor noise, rotation direction and function										
Check pump suction										
Check pump sieves correctly fitting and operating correctly										
Check sliding ring washer/contra-rotation ring										
2. Washing systems										
Check water level in tank										
Check that wash water pipe is watertight										
Check washing system is complete and produces correct spray pattern										
Check wash arm hubs										
3. Fresh water rinse										
Check flow pressure/water pressure										
Check rinsing system is complete and produces correct spray pattern										
Check that system is watertight										
4. Housing and mounting parts										
Check housing, tank, sheet metal cover, hood, doors and covering of machine base for damage and correct operation										
Check tank cover sieves										
Check boiler, hoses, clamps, plastic parts and seals										
Check operation of raising and lowering equipment										
5. Fresh water installation										
Check level regulation										
Check valves, clean dirt trap										
Check that all connections (incl. hand spray) are watertight										
Check settings of built-in water softener (if fitted)										
Check operation of complete or partial water softener (if fitted)										
Check water hardness										
6. Waste water installation										
Check if watertight										
Check pressure hose position and operation of wash water evacuation pump										

Service plan for stationary rack machines

Service procedures	FV 28G / FV28GIO EcoStar 430 F / 530 F EcoStar 530 F-M	FV 40.2 / FV 60.2 / FV 70.2 D	GK 60	OR 50 H	DV 40 – DV 240 B EcoStar 545D / DV 80.2 / DV 120.2 / DV 200.2 / DV 200.2 PW	DV 270 B	FV 130.2 – FV 250.2 DV 270.2	Component OK	Component faulty	Component replaced
7. Electrical installation										
Check of all fuses										
Tighten all electrical connections										
Check tank and boiler heating										
Check thermostat and stop switch										
8. Detergent dosage										
Check dosage, adjust if necessary										
9. Rinsing dosage										
Check dosage, adjust if necessary										
10. Operation check of the complete machine										
Check stationary rack machine for correct interaction of all operations										
11. Test run										
Check results of test wash and rinse										
Brief instruction for new personnel										

Saugleitung für Reiniger bzw. Klarspüler
Suction line for detergent resp. rinse aid
Conduite d'aspiration pour détergent, resp. produit tensio-actif
Tubo d´aspirazione per detergente e brillantante
Tubo de aspiración para detergente y abrillantador



ACHTUNG!	**Saugleitung von Wärmequellen fernhalten!**
ATTENTION!	**Keep away suction line from heating sources!**
ATTENTION!	**Ecartez la conduite d'aspiration de toute source de chaleur!**
ATTENZIONE!	**Tenere il tubo d´aspirazione lontano da fonti di calore!**
¡ATENCION!	**¡Mantenga el tubo de aspiración alejado de las fuentes de calor!**

CE - Konformitätserklärung

gemäß EN 45014 und
EG-Maschinenrichtlinie 98/37/EG
Stand: 29.09.2005

CE declaration of conformity as defined by EC machinery-directive
Déclaration de conformité CE conformément à la directive CE relative aux machines
Declaración de cónformidad CE según los requerimientos CE en la construcción de maquinas CEN03A/01/98
CE-Conformiteitsverklaring volgens de EG Machinerichtlijn
CE – dichiarazione di conformitá secondo le direttive stabilite riguardo alla costruzione di macchine

Firma/Company/Société/Empresa/Firma/Casa costruttrice:
Adresse/Adress/Adresse/Dirección/Adres/Indirizzo:

MEIKO Maschinenbau GmbH & Co. KG
Englerstraße 3
D-77652 Offenburg
e-mail: info@meiko.de

Spülmaschine Typ	**FV 28G**	**FV 110G**	**DV 40N**	**DV 120.2**	**EcoStar 430F**	**OR 50H**
dishwashing model /	**FV 20N**	**FV 130B**	**DV 40T**	**DV 120T**	**EcoStar 530F**	**GK 60**
lave-vaiselle modèle /	**FV 40T**	**FV 250B**	**DV 80T**	**DV 160**		
lavastoviglie modello	**FV 40.2**	**FV 130.2**	**DV 80.2**	**DV 200.2**	**EcoStar 545D**	
	FV 60.2	**FV 250.2**		**DV 200.2 PW**		
	FV 70.2			**DV 240B**		
	FV 70T			**DV 270B**		
				DV 270.2		

Konformitätserklärung/Declaration of conformity/Déclaration de conformité/Declaración de conformidad/Conformiteitsverklaring/ Dichiarazione di conformitá:

Hiermit bescheinigen wir in alleiniger Verantwortung die Konformität des Erzeugnisses mit den grundlegenden Anforderungen der folgenden EG-Richtlinien, harmonisierten Normen, nationalen Normen.

We herewith confirm the sole responsibility for the conformity of the product with the basic requirements of the following EC-regulations, harmonized standards, national standards.

Par la présente nous déclarons, que nous avons responsabilité pour la conformité du produit aux demandes fondamentales des régulations CE, normes harmonisées et normes nationales suivantes.

Por la presente atestamos en exclusiva responsabilidad la conformidad de nuestros productos con los requerimientos básicos de los siguientes requerimientos CE, normas armonizadas y nacionales.

Hierniee bevestigen wij onze verantwoordelijkheid van de conformiteit van het product met betrekking tot de fundamentele en gestelde eisen volgens EG-Richtlijnen, geharmoniseerde Normen en Nationale Normen.

Con la presente dichiarazione confermiamo la nostra responsabilità riguardo alla conformitá sul prodotto con i regolamenti basilari delle seguenti normative CE, normative armonizzate e normative nazionali.

EG-Richtlinie/EC-regulation/Régulation CE/Requerimiento CE/EG-Richtlijn/Regolamento CE:
98/37 EWG

Die Geräte sind gebaut unter Berücksichtigung folgender Normen:

The devices are constructed under consideration of the following standards / Gli apparecchi sono realizzati nel rispetto delle seguenti norme / Los aparatos han sido iseñados considerando las siguientes normas / Les appareils ont été conçus en respectant les normes suivantes

Harmonisierte Normen/Harmonized standards/Normes harmonisées/Normas armonizadas/Geharmoniseerde Normen/normative armonizzate:
EN 60335-1 / EN 60335-2-58 / EN 294 / EN 349 / EN 563 / EN 954-1 / EN 61000-6-2 / EN 61000-6-3 / pr EN 13621 (nur für Salatschleuder)

Nationale Normen/National standards/Normes nationales/Normas nacionales/Nationale Normen/narmative nazionali:
DIN EN 12100-1 / DIN EN 12100-2 / DIN 1717 / DVGW-W 507
DIN 10511 (gilt für Gläser-Geschirrspülmaschinen) / DIN 10512 (gültig für Eintank-Geschirrspülmaschinen) / DIN 10522 (gültig für Behälter-Spülmaschinen).

VGG-Merkblätter/VGG-leaflets/VGG-feuilles informatives/VGG-hoja informativa/VGG-Datablad/VGG schede informative

Offenburg, den 11.10.2005
Offenburg, the/Offenburg, le/Offenburg, el/Offenburg,/Offenburg, il

Unterschrift/Signature/Signature/Firma/Handtekening/firma:
Konstruktion/Construction/Construction/Construcción/Constructie/resp. progettazione:

ppa